

Disproof of conjecture 00000002531

The stated density equals two on the real line

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Claim and conventions. Both language versions define the density by

$$\theta_E(x) = \lim_{r \downarrow 0} \frac{\mathcal{H}^k(E \cap B(x, r))}{r^k}$$

and assert that every $\mathcal{H}^k$ -measurable set has, outside an $\mathcal{H}^k$ -null set, a density limit in $\{0, 1\}$. They also assert dimension and sharpness bounds for the set of points where the limit fails to exist. The denominator above is exactly the source's r^k .

The source leaves the ambient space and normalization implicit. We use the ordinary Euclidean interpretation, positive integer dimension, open balls centered at x , and all positive real radii approaching zero. Standard normalized Hausdorff measure is $\omega_k/2^k$ times the diameter-cover measure, where ω_k is the volume of the Euclidean unit ball [1]. We take $k = n = 1$. Then $\omega_1 = 2$, so the coefficient is one, and $\mathcal{H}^1$ on $\mathbb{R}$ is ordinary length. These conventions are explicit in the Lean definitions. No unit-ball factor is inserted into the source's density denominator.

Proposition 1. *Let $E = \mathbb{R}$ with its usual distance $d(x, y) = |x - y|$. With the stated density formula and standard $\mathcal{H}^1$, the density exists and equals 2 at every $x \in \mathbb{R}$. No $\mathcal{H}^1$ -null exceptional set makes this density binary.*

Proof. The set E is Borel and hence $\mathcal{H}^1$ -measurable. For every $x \in \mathbb{R}$ and every $r > 0$,

$$E \cap B(x, r) = (x - r, x + r), \quad \mathcal{H}^1(E \cap B(x, r)) = 2r, \quad \frac{\mathcal{H}^1(E \cap B(x, r))}{r^1} = 2.$$

Thus the full positive-radius limit is 2, which is neither 0 nor 1. If a null set N made the binary conclusion true for every $x \in E \setminus N$, then $E \setminus N$ would be empty. This forces $N = \mathbb{R}$, contradicting $\mathcal{H}^1(N) = 0$, since $\mathcal{H}^1(\mathbb{R}) = \infty$. $\square$

Consequence and scope. The first necessary assertion is false, so the conjunction in the conjecture is false. This example has a limit everywhere; it does not separately refute the later dimension or sharpness assertions. The source imposes no finite total measure hypothesis. Although $\mathcal{H}^1(E) = \infty$, every ball numerator in the displayed calculation is finite. The disproof applies whether “almost everywhere” is read on E or on the ambient space, since here $E = \mathbb{R}$. It depends on the stated normalization and denominator; replacing the denominator by $\omega_k r^k$ would be a different assertion.

Correspondence with Lean. The project pins Lean 4.19.0 and Mathlib v4.19.0. All mathematics is in `Conjecture2531.lean`, under the namespace `Conjecture2531`. The genuine library measure `Measure.hausdorffMeasure 1` is multiplied by the explicitly defined coefficient $\omega_1/2^1$. Theorems prove this coefficient is one and the resulting measure equals `volume`, using Mathlib’s `hausdorffMeasure_real`. The genuine real metric, ball measure, finite ball measures, and infinite total measure are also checked.

The function `densityQuotient` divides the ball-intersection measure by the extended nonnegative real value of r^1 . Its extended codomain preserves infinity. For $r > 0$, the denominator is positive and finite.

`HasBinaryDensity` asserts convergence to a value equal to zero or one along `nhdsWithin 0 (Set.Ioi 0)`. The radius filter is proved nontrivial; the theorem uses uniqueness of limits, with the usual topology on the extended nonnegative reals.

`NullExceptionalBinaryDensity` quantifies an arbitrary null set N for each E and requires binary density at every point of $E \setminus N$. For Borel E , a theorem identifies this with the restricted-measure almost-everywhere formulation, without assuming the density predicate is measurable. A separate theorem shows that ambient almost-everywhere binary density implies this same necessary assertion.

The definition `HausdorffMeasurable` uses the actual Carathéodory predicate of the Hausdorff outer measure. Borel measurability implies it. The formal propositions `BorelDensityClause` and `CaratheodoryDensityClause` are explicitly necessary specializations of the first source assertion to $k = n = 1$. They are not definitions of the entire compound conjecture. The main results are:

- `counterexample`: $\mathbb{R}$ is measurable in both senses, its density converges to two everywhere, and no null exception restores binary density;
- `not_borelDensityClause` and `not_caratheodoryDensityClause`: negations of the necessary clauses;
- `not_statement_implying_borelDensityClause`: a logical bridge from any explicitly supplied necessary-clause implication to its negation;
- `not_borelDensityClause_and`: further conjunctive assertions cannot rescue the false clause; those assertions remain uninterpreted.

Verification. A fresh independent build and source replay with warnings treated as errors passed. The audit covers 31 authored declarations (nine definitions and 22 theorems) and six generated declarations, including their logical dependencies. There are no admitted proofs, unsafe or partial dependencies, custom axioms, or `native_decide` calls. All axioms belong to the standard set: `propext`, `Classical.choice`, `Quot.sound`. The auxiliary inspection replay passed. No separate numerical experiment is required. Detailed records and reproduction instructions are included; local verification does not constitute maintainer acceptance.

References

- [1] Leon Simon, *Introduction to Geometric Measure Theory*, Chapter 1, §2 (Hausdorff measure normalization), author-hosted text. The source conjecture’s density denominator is retained independently.
- [2] Mathlib v4.19.0, `Mathlib.MeasureTheory.Measure.Hausdorff` and the real Lebesgue measure library; exact revision is pinned in `lake-manifest.json`.